TCS CORPORATE WEBSITE

Image Collage Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

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Image Collage Block — Developer Guide

Attribute	Value
Document ID	DEV-B-image-collage
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Image Collage block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/image-collage/ and its components image-collage, image-collage-item
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/image-collage/README.md, blocks/image-collage/_image-collage.json, blocks/image-collage/image-collage.css, blocks/image-collage/image-collage.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	—

1. Summary

Attribute	Value
Block	Image Collage
Folder	`blocks/image-collage`
Type	Container
Category	content
Status	Active (Proposed, KB-D-063)
Purpose	Centred eyebrow, title, description and CTAs on a dark background framed by up to nine floating decorative images
Variations (summary)	Title weight, description size, eyebrow style, CTA styles, backgrounds and shapes, image treatment
Children	image-collage-item (Collage image item)
Code size	1366 lines of JS and CSS

2. Overview

- **Business purpose:** a visual statement section, for example to introduce a portfolio, a story or a brand theme, where many small images set the mood around one message.
- **User experience:** a dark block at least 64rem high (46.875rem on tablets) with a soft overlay over the images. The text column is centred and at most 37.5rem wide. The images float slowly (not with reduced motion) and are decorative (`aria-hidden`), so screen readers read only the text and the CTAs.
- **Templates / page types:** any page. No template adds the block.

- **Data source:** authored fields and the Collage image items only. Placeholders: `opensInNewTab` (CTAs opening a new tab) and `itemEditorHint` (empty items in Universal Editor).
- **Related blocks:** Teaser (`blocks/teaser`), whose Layout Image collage has the same purpose (see KB-D-063); Hero (`blocks/hero`) for the opening banner of a page.
- **When to use:** one message framed by a set of images, with at most two CTAs.
- **When not to use:** when the images carry information (they are decorative and hidden from assistive technology; use Cards or a List); for the page's opening banner (use a Hero).
- **Realistic use cases:** a "Our work" introduction above a case study list; a brand campaign section; an events or culture page section.

3. Source files

File	Link
`README.md`	GitHub
`_image-collage.json`	GitHub
`image-collage.css`	GitHub
`image-collage.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`image-collage`	Image Collage	`core/franklin/components/block/v1/block`	Image Collage	`image-collage`	image-collage-item
`image-collage-item`	Collage image item	`core/franklin/components/block/v1/block/item`	Collage image item	`image-collage-item`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Image Collage (`image-collage`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Title	`title`	text	No	—	—	—
Content	Title Type	`titleType`	select	No	h2	6 options	—
Content	Hide Title	`titleHidden`	boolean	No	`false`	—	—
Content	Description	`description`	richtext	No	`""`	—	—
Content	Show Description	`showDescription`	boolean	No	`true`	—	—
Content	Primary CTA Title	`primaryCtaText`	text	No	—	—	—
Content	Primary CTA Link	`primaryCta`	aem-content	No	—	—	—

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Primary CTA Link Type	`primaryCtaTarget`	select	No	`""`	2 options	—
Content	Primary CTA Link Style	`primaryCtaStyle`	select	No	`primary`	4 options	—
Content	Primary CTA Background	`primaryCtaBackground`	select	No	`""`	4 options	—
Content	Primary CTA Shape	`primaryCtaShape`	select	No	`""`	2 options	—
Content	Secondary CTA Title	`secondaryCtaText`	text	No	—	—	—
Content	Secondary CTA Link	`secondaryCta`	aem-content	No	—	—	—
Content	Secondary CTA Link Type	`secondaryCtaTarget`	select	No	`""`	2 options	—
Content	Secondary CTA Link Style	`secondaryCtaStyle`	select	No	`secondary`	4 options	—
Content	Secondary CTA Background	`secondaryCtaBackground`	select	No	`""`	4 options	—
Content	Secondary CTA Shape	`secondaryCtaShape`	select	No	`""`	2 options	—
Style	Title weight	`titleWeight`	select	No	`""`	3 options	—
Style	Description size	`descriptionSize`	select	No	`""`	3 options	—
Style	Background color	`backgroundColor`	select	No	`""`	2 options	—

5.2 Collage image item (`image-collage-item`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Collage image	Image	`image`	reference	Yes	—	—	—
Collage image	Image alt text	`imageAlt`	text	No	—	—	—
Collage image	Image treatment	`imageTreatment`	select	No	`black-white`	2 options	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../scripts/aem.js`
`../../scripts/scripts.js`
`../../decorators/a11y-decorator.js`
`../../decorators/button-decorator.js`
`../../decorators/content-decorator.js`
`../../decorators/editor-item-decorator.js`

Import
`../../decorators/heading-decorator.js`
`../../decorators/image-decorator.js`
`../../utils/cells.js`
`../../utils/links.js`
`../../utils/placeholders.js`
`../../utils/properties.js`

7. For Developers

```
blocks/image-collage/
├── README.md
├── _image-collage.json
├── image-collage.css
└── image-collage.js
```

Decorator/Utility	Purpose	Execution or dependency point
scripts/aem.js	Edge Delivery core library (block loading, CSS, metadata, optimized pictures)	`image-collage.js`: { toClassName }
scripts/scripts.js	Page decoration (moveInstrumentation and shared page helpers)	`image-collage.js`: { moveInstrumentation }
decorators/a11y-decorator.js	`hideVisually`, `createHiddenText` (with `.visually-hidden` in `styles/styles.css`)	`image-collage.js`: { hideVisually }
decorators/button-decorator.js	`decorateButton` (option `newTabHint`: a link opening a new tab says so), `createButton`, `setButtonDisabled`, `setButtonAlignment`, `removeButtonDecoration`	`image-collage.js`: { decorateButton }
decorators/content-decorator.js	`fromCell` (property content into an element, keeping editor instrumentation)	`image-collage.js`: { fromCell }
decorators/editor-item-decorator.js	`buildEditorItems`, `editorItemHint` (in Universal Editor, an item the block doesn't render yet, e.g. a CTA just added, as a dashed placeholder the author can select; styles in `styles/lazy-styles.css`; the hint loaded o	`image-collage.js`: { buildEditorItems, editorItemHint, }
decorators/heading-decorator.js	`buildHeading` (Title + Title Type), `HEADINGS`	`image-collage.js`: { buildHeading }
decorators/image-decorator.js	`pictureOf` (image with its alt text field)	`image-collage.js`: { pictureOf }
utils/cells.js	`itemRows`, `KEY_VALUE_ITEM`, `FIELD_ROW_ITEM` (the item rows of a block: by model in the editor, by their cells on the page), `withoutTabCells` (drops the empty cell AEM renders for each dialog tab of an item model), `m	`image-collage.js`: { KEY_VALUE_ITEM, itemRows }
utils/links.js	`NEW_WINDOW`, `linkOf`, `linkPathsOf` (the paths of a page property, one or many), `readLinkItems` (CTA lists with Link Type and the items' other select values), `readLinkRows` (the same from link item rows, e.g. Hero CT	`image-collage.js`: { NEW_WINDOW, linkOf }
utils/placeholders.js	`getPlaceholders`, `loadPlaceholders`, `fillPlaceholder` (UI text from the locale's placeholders, with English fallbacks)	`image-collage.js`: { getPlaceholders }
utils/properties.js	`readProperties`, `textOf`, `isOn`, `isOnByDefault` (a toggle that defaults to on), `optionOf`, `normalizeKey`, `textSource` (key-value block properties)	`image-collage.js`: { isOn, isOnByDefault, optionOf as option, readProperties, textOf, }

- **JavaScript (`image-collage.js`):** an async `decorate(block)`. It reads the rows with `readProperties`, finds the Collage image items with `itemRows(block, ['image-collage-item'], KEY_VALUE_ITEM)` (by model in Universal Editor, by their cells on the page), loads the placeholders, then builds the media layer, the content and the editor placeholders and replaces the block's children.
- **CSS (`image-collage.css`):** the dark block and its overlay (`::before`), the content column, the nine image slots and their float animation, the CTA styles, and the three breakpoints.
- **Input DOM:** key-value rows `<div><div>name</div><div>value</div></div>`, then one row per Collage image item with three cells (the dialog tab, the image with its alt text, the treatment).
- **Rendered DOM:**

```
<div class="image-collage block title-semibold">
  <div class="image-collage-media" aria-hidden="true">
    <div class="image-collage-item image-collage-item-1 image-black-white"><picture>…</picture></div>
    …
  </div>
  <div class="image-collage-content">
    <div class="image-collage-intro">
      <div class="image-collage-heading-row"><h2 class="image-collage-title">…</h2></div>
      <div class="image-collage-description">…</div>
    </div>
    <div class="image-collage-links">
      <a class="button primary image-collage-cta">…</a>
      <a class="button secondary image-collage-cta">…</a>
    </div>
  </div>
</div>
```

- **Block classes set by the code:** the Style options (`title-regular`, `title-semibold`, `text-regular`, `text-large`, `bg-navy`), removed first and re-added on each decoration.
- **Events and async behavior:** one `getPlaceholders()` request (cached per page).
- **Error handling:** no try / catch; missing cells give empty parts, items without an image aren't shown.
- **Editor:** `moveInstrumentation`, `fromCell` and `buildHeading` keep Universal Editor's instrumentation on the rendered elements; items not shown become placeholders (`buildEditorItems`).
- **Motion:** `decorators/motion-decorator.js` animates the content column (`'image-collage': { targets: ':scope > .image-collage-content' }`) and treats the media layer as the background (`BACKGROUNDS['image-collage']`).

8. Extending the Block

1. Add a field in `_image-collage.json` (at most three short tabs, Additional last), then run `npm run build:json` and `npm run lint`.
2. Follow the AEM delivery rules (field collapse, no grouped fields, `classes_*` for classes, no sample content in the template).
3. Read it in `image-collage.js` and style it in `image-collage.css`, scoped to `.image-collage`, with rem sizes.
4. Run `npm run build:modules` after changing imports and `npm run build:docs` after changing the model.

9. Responsive Behavior

- **Mobile (< 600px):** the images keep their positions in a fixed 64rem square composition, centred and cropped at the sides; title 2rem; CTAs 3.125rem high with 0.75rem text.
- **Tablet (600px to 1199px):** the block is at least 46.875rem high; title 2.5rem; CTAs 3rem high with 0.875rem text.
- **Desktop (1200px and above):** the images are placed in percentages of the block; title 3rem; CTAs 3rem high with 1rem text.

10. Accessibility

- The images are in an `aria-hidden` layer: only the text and the CTAs are announced.
- The title is a real heading of the authored level; Title hidden keeps it for screen readers.
- CTAs opening a new tab announce it (`opensInNewTab` placeholder, `announceNewTab` through `decorateButton`).
- The floating motion stops with `prefers-reduced-motion: reduce`.
- Check the contrast of the description (`#9e9fa3`-like grey on the dark background) and of any colour images behind the text.

11. Data Layer and Analytics

The block is a component entry of the Adobe Client Data Layer like every block (`decorators/data-layer-decorator.js`); clicks on its CTAs are captured centrally. The block pushes nothing itself.

12. Performance Considerations

- Up to nine images per block: keep them small (they are shown at most a few hundred pixels wide) and use AEM Assets renditions.
- In the first section the block can hold the Largest Contentful Paint; prefer it below the fold.
- The float animation runs on `transform` only and stops with reduced motion.

13. Security and Privacy

Authored content only. No external HTML is injected, no personal data is read or pushed to the data layer.

14. For Architects

- **Role:** a presentational, authored-content block with item children; no data source.
- **Overlap:** the Teaser's Image collage layout covers the same design; decision KB-D-063 (Proposed) records whether both stay.
- **Dependencies:** `decorators/a11y-decorator.js`, `button-decorator.js`, `content-decorator.js`, `editor-item-decorator.js`, `heading-decorator.js`, `image-decorator.js`; `utils/cells.js`, `links.js`, `placeholders.js`, `properties.js`.
- **Performance:** one cached placeholders request; images are lazy unless the block is in the first section.

15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
No images on the page	Items without an image, or images not published	Inspect `.image-collage-media`	Select the images and publish them with the page	The media layer has items
An image is missing	More than nine items	Count the items	Remove the extra items	Nine images

16. Known limitations of the block

- At most nine images; the slot positions are fixed by the design.
- The images are decorative only (`aria-hidden`); their alt text isn't announced.
- The Teaser's Layout Image collage offers the same design (KB-D-063, Proposed).

- The colours are fixed values of the design (`--image-collage-*` custom properties), not the site's semantic tokens, so a dark Section's theme contract doesn't change them.

17. Change history of the block

Version	Change	Source	Date
Not recorded	New block (model, code and README)	Repository	29 October 2026

18. Reference pages

Kind	Page	Live URL

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	IMAGE-COLLAGE-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	IMAGE-COLLAGE-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

> **Screenshot required**
>
> - Required screen: Image Collage on its block library page
> - Required role/access: none (published page)
> - Required navigation steps: Open https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/image-collage

> - Reason unavailable: No block library page of this block was found under /design-block-master/en/block-library
> - Owner: Content Lead / AEM Authoring Lead (proposed)
> - Target completion date: To be confirmed

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages